Problem Chosen	2026 HiMCM Summary Sheet	Team Control Number
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Choosing a City for High-School Life with Incomplete Evidence

Summary

Zurich ranks first among the sixteen cities, scoring 64.5, followed by Madrid (63.3) and Singapore (62.5). We use AHP to turn a high schooler's stated priorities into category weights: 0.340 each for safety and education, 0.123 each for cost and mobility, and 0.073 for daily life. Equal weights give the same 0.10 weight to every indicator. TOPSIS combines the ten utility scores, using the midpoint of each known range for the main ranking. The weighting change moves 12 of 16 cities; the leading four shift from Zurich, Oslo, Paris and Dublin to Zurich, Madrid, Singapore and Milan.

Zurich also has the smallest worst-case regret, 6.1 points. Of the 160 city-indicator entries, 123 contain a single value, including proxies, and 37 contain known ranges; nine cities could rank first if their ranges broke in their favor. Zurich remains first in all 31 controlled source cases and in 82.8% of 10,000 weight perturbations at concentration 100. Removing its homicide input makes Madrid first. The three 2035 scenarios leave the top three unchanged. Doubling the education or mobility weights separately also leaves Zurich first. Country, district and age-group cases distinguish facility supply from student access and broad observed groups from unsupported age-specific preferences.

Keywords: high-school city choice; AHP; TOPSIS; interval evidence; maximum regret; student priorities.

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1 Introduction and decision scope

1.1 What makes a city suitable for high-school life?

Zurich leads the sixteen-city comparison with a TOPSIS score of 64.5, followed by Madrid (63.3) and Singapore (62.5). A useful comparison links a student's school day to safety, air quality, teaching resources, housing and living costs, transport, broadband and swimming facilities. We define "best" as the highest score under AHP weights based on the team's high-schooler viewpoint; known utility ranges enter at their midpoints. A separate worst-case regret check measures each city's largest possible score shortfall. Zurich has the smallest, at 6.1 points. This is a starting point among the sixteen candidates; families still choose according to a student's needs.

The comparison uses the frozen team matrix of sixteen cities, one per country, exceeding the ten-city minimum and spanning different resource and cost settings. This inherited set is not a representative global sample. Findings concern these candidates; broader claims require an explicit sampling design and additional comparable observations.

1.2 Criterion selection and contribution

The ten indicators preserve the agreed team framework and balance education with the conditions that make resources usable in daily life. Their directions and anchors appear in Table 2. Counts and city averages serve as scoped supply or setting measures. They do not directly measure friendships, student wellbeing, school quality, language fit, discrimination, cultural participation or individual access. Those outcomes are relevant to a family's choice; comparable student-level observations are needed to integrate them without inventing measurements or double-counting existing proxies.

Our main revision replaces equal indicator weights with an AHP hierarchy based on the team's high-schooler viewpoint. Equal weights give the same 0.10 to each measure, so a pool count carries as much weight as classroom resources. AHP compares five categories, then the two indicators within each. Pupils per teacher receives 0.255 overall weight; pools receive 0.024. Equal weights place Zurich, Oslo, Paris and Dublin in the top four; AHP places Zurich, Madrid, Singapore and Milan there. Twelve ranks change ($\tau = 0.517$). TOPSIS combines these weights with the ten utilities. Source cases, weight perturbations, preference changes and 2035 scenarios test the recommendation; VIKOR gives one focused comparison.

AI assistance supported research retrieval, calculations, code and prose preparation, including Exa source retrieval and extraction [19, 20]. The appended use report records the known assistance and the outstanding documentation required before submission.

2 Assumptions and their role

1. **Comparability follows the retained scope.** Original values and population denominators remain fixed. Each supplement keeps its provider, year, city boundary and educational population. Mixed-year densities are conditional comparisons; they do not create a same-year census.

2. **Utilities are monotone on fixed scales.** More resources and less cost or burden improve their respective utilities. Declared anchors set the scale; they are neither estimated preferences nor empirical optimum values.
3. **Weights express a stated viewpoint.** AHP translates the team's high-schooler priorities into weights: 0.340 each for safety and education, 0.123 each for cost and mobility, and 0.073 for daily life. These are stated priorities, not estimates from a student survey.
4. **The score permits compensation.** Weighted distances combine criteria in a common utility space. The model therefore allows stronger results on some criteria to offset weaker results on others. A family requiring a hard safety or eligibility threshold should apply it separately before choosing.
5. **The evidence set is rectangular.** Of 160 entries, 123 contain a single value, including proxies, and 37 contain known ranges. We vary those ranges independently, omitting cross-criterion and cross-city restrictions. Nine cities could rank first if their ranges favor them; these champions and regret bounds are conditional on this set, not population probabilities.
6. **Extensions hold the comparison basis fixed.** Scenario changes retain population denominators, anchors and the indicated weight rule. They are conditional experiments with no fitted future date or representative age-specific preference distribution.

3 Notation

Symbol	Meaning
x, a	Observed indicator value and reference level (in the same units)
u, u_{ij}	Utility on $[0, 1]$; i identifies a city and j an indicator
c, P, d	Facility count, city population and facilities per 100,000 people
t, q	Travel duration in minutes and its utility
w_j	Indicator weight, with all ten weights summing to one
D_i^+, D_i^-	City i's distances to the ideal and worst utility profiles
S_i	City i's TOPSIS score, on $[0, 100]$
ℓ_{ij}, h_{ij}	Lower and upper utility bounds
L_i, U_i	Lower and upper score bounds, $S(\ell_i)$ and $S(h_i)$
R_i	Maximum regret
κ	Concentration of a declared Dirichlet weight distribution
r	Raw-value multiplier in a development scenario

4 Indicator construction and city selection

4.1 City selection

The sixteen cities are Amsterdam, Berlin, Dublin, Helsinki, London, Madrid, Milan, Oslo, Paris, Prague, Singapore, Stockholm, Tokyo, Vienna, Warsaw and Zurich. The primary

comparison is a set of sixteen cities in sixteen countries, not a convenience sample. Four selection rules determine membership, and we state them here so that the scope of every later claim is visible rather than assumed.

R1 — One city per country. Sixteen cities, sixteen countries, so no national data system, qualification structure, or policy regime is counted twice. A country's statistical conventions would otherwise leak into several rows at once and tilt the ranking; the one-city rule keeps the national effect and the city effect from being entangled in the primary comparison, and it leaves the within-country check of Section 7 as a separate experiment rather than a confound inside the ranking.

R2 — Comparable, city-level official data. Each of the ten indicators must be measurable at city level from an official statistical office, transport operator, or municipal registry. The sixteen cities all publish the required records—homicide statistics, $PM_{2.5}$ monitoring, price-to-income, school staffing, facility inventories, and transport-stop registries—in forms that can be reconciled to a common unit. This is the decisive constraint. A city that can supply a defensible value for only a few indicators would inject unknown rather than bounded uncertainty, so we exclude it from the primary set instead of papering over the gap with a proxy. The whole of Section 4 is built on this requirement.

R3 — Controlled variation on the measured dimensions. The set is not uniform. It spans the ranges that the ten indicators are meant to separate, so the ranking is a genuine comparison rather than a coronation of a single profile. Table 1 summarizes the spread.

Table 1: The sixteen cities distribute across the dimensions the indicators measure.

Dimension	Span within the set
Cost of living	Moderate capitals (Warsaw, Prague, Madrid, Berlin) to among the world's most expensive (Zurich, Oslo, London, Singapore)
Climate	Cold northern (Oslo, Helsinki, Stockholm) and continental (Berlin, Vienna, Warsaw, Prague) through maritime (London, Dublin, Amsterdam) and warm Mediterranean (Madrid, Milan) to tropical Singapore and humid-subtropical Tokyo
Scale	Mid-size capitals (Zurich, Oslo, Helsinki, Dublin) up to large metropolises (Tokyo, London, Paris)
Mobility	Cycling and tram networks (Amsterdam, Berlin, Prague, Vienna) to the densest metro systems (Tokyo, Paris, London)
Safety	The lowest recorded homicide rates (Singapore, Tokyo, Oslo) up to the higher European rates elsewhere in the set

R4 — Above the minimum, with every cell traceable. The problem asks for at least ten cities; sixteen supplies six more than the minimum while keeping all 16×10 cells indi-

vidually auditable. The margin matters less than the traceability: the interval analysis of Section 5.4 can only treat an unknown cell honestly if the set is small enough to document.

What the set is not. Fourteen of the sixteen cities are European and every city sits in a high-income economy, so this is a deliberately comparable high-income comparison group, not a representative sample of world cities. We say this plainly rather than hiding it. The consequence is exact: the recommendation of Section 5 is a statement about these sixteen cities, and no conclusion transfers to a city outside the set without an explicit sampling design. That is the same honesty Section 2 applies to the indicators, and it is why every headline result in this paper is phrased "among the sixteen cities."

4.2 Indicator system

We group the ten indicators into five parts of high-school life (Table 2). Each group contains two indicators and forms one category in the weight hierarchy. A lower burden or a greater supply gives a higher utility, so all ten indicators can be compared on a scale from 0 to 1.

Table 2: The ten indicators and their role in a high schooler's daily life.

Category	Indicator	Unit	Better when	Why it matters for a high schooler
Safety and environment	Homicide rate	Deaths per 100,000	Lower	Personal safety
	$PM_{2.5}$ concentration	$\mu g/m^3$	Lower	Air breathed each day
Cost of living	Living-cost index	Index points	Lower	Family spending
	Housing price-to-income ratio	Ratio	Lower	Housing affordability
Education	Upper-secondary pupils per teacher	Pupils per teacher FTE	Lower	Teaching resources
	Higher-education institutions	Institutions per 100,000	Higher	Options after school
Mobility	Mean travel/commute time	Minutes	Lower	Time left after travel
	Public-transport stops	Stops per 100,000	Higher	Transport supply
Daily life	Fixed broadband speed	Mbps	Higher	Study and communication
	Public swimming pools	Venues per 100,000	Higher	Exercise opportunities

Putting different units on one scale. For a benefit indicator, such as broadband speed, utility is

$$u = \frac{x}{a + x}. \tag{1}$$

For a cost indicator, such as homicide rate, utility is

$$u = \frac{a}{a + x}. \tag{2}$$

In both formulas, the anchor a gives utility 0.5 when $x = a$. For homicide and $PM_{2.5}$, the anchors are 2 deaths per 100,000 and 15 $\mu g/m^3$. Living costs and housing use 100 index points and a price-to-income ratio of 10. The education anchors are 15 pupils per teacher and 1 higher-education institution per 100,000. Transport stops use 50 per 100,000; broadband uses 100 Mbps and pools use 1 venue per 100,000. These reference levels stay fixed across cities.

Facility supply per resident. For transport stops and swimming pools, we convert counts to density per 100,000 people:

$$d = \frac{100000c}{P}. \tag{3}$$

We then use $x = d$ in Equation 1. A verified subset of facilities gives a lower utility bound; the upper bound stays at 1 when other providers remain uncounted. Density measures supply across a city, so a high value does not guarantee a short walk from every student's home.

School resources at the right stage. We measure upper-secondary teaching resources as pupils per full-time-equivalent (FTE) teacher:

$$x = \frac{\text{upper-secondary pupils}}{\text{upper-secondary teacher FTE}}. \tag{4}$$

Prague's 2024/25 records give 71,325 upper-stage pupils and about 6,139 teacher FTE after removing lower-gymnasium years, or about 11.6 pupils per teacher [11]. This measures staffing resources available to schools.

Time spent travelling. A longer journey leaves less time for homework and activities. We use

$$q = \exp(-t/30). \tag{5}$$

The 30-minute scale gives a half-hour journey utility of about 0.37. When only a mean duration $\bar{t}$ is known, the average travel utility lies in

$$\mathbb{E}[q] \in [\exp(-\bar{t}/30), 1]. \tag{6}$$

The curve is convex, so its value at the mean gives a lower bound on average utility; 1 is the upper limit for nonnegative durations. The retained work-commute means describe

general travel conditions, while Singapore's school-stage travel bins give a separate range [56, 14].

Figure 1 compares the cities on the shared utility scale. Singapore and Milan lead on homicide utility at 0.97 and 0.89, followed by Zurich at 0.82; Amsterdam is lowest at 0.06. Among the pupil-teacher proxies, Madrid has the lowest ratio at 9.6 pupils per teacher, with utility 0.61.

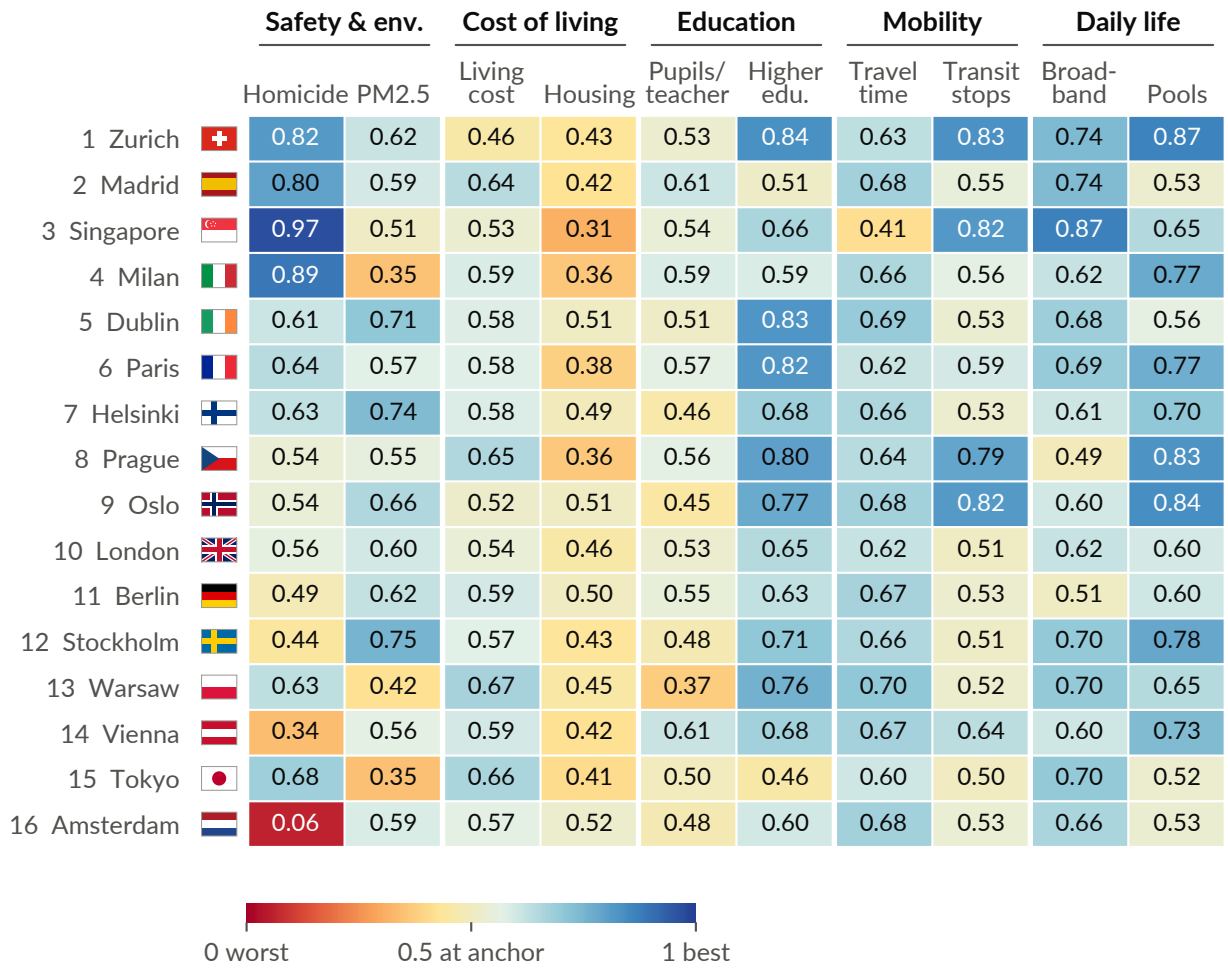


Figure 1: Indicator utilities for the sixteen cities, ordered by final score and grouped by category. Colour runs from 0 (worst) to 1 (best); a cell with a known range shows its midpoint. Sources: team table, official supplements and stated national or survey proxies.

4.3 Data sources and filled values

The team table is our starting matrix [1]. We keep its supplied values, populations and source years. Official statistical tables, operator records, municipal sources and stated proxies supplement the matrix. We retain each source's year and geographic scope, and

identify entries drawn from different years. Supplied values without original source links remain attributed to the team table. Their influence is checked in Section 5.5.

Three rules keep the measurements comparable:

1. Count each station once, excluding separate platforms and repeated line entries; Warsaw's 21 M1 and 18 M2 entries give 38 stations after merging the shared Świętokrzyska interchange [58].
2. Match school stage before dividing pupils by teacher FTE; Prague's totals require removing lower-gymnasium pupils and their teachers [11].
3. Replace Zurich's unverified supplied rate of 0.14 with about 0.45 homicides per 100,000, calculated as two completed homicide offences in 2025 divided by its 2025 population of 440,935 and scaled to 100,000 residents [41].

Tokyo's Oedo ticket-service directory lists 38 distinct stations in the 23 wards, and we do not add entries distinguished only by machine-readable flags. [57] Oslo's bus-stop inventory uses parent stop groups [6]. We exclude the questioned Tveten bru entry as a precaution, its closure being unconfirmed.

Eurostat table educ_uoe_perp04 (ISCED 3, 2024) supplies national pupil-teacher ratios for Helsinki 17.4, Amsterdam 16.4, Dublin 14.2, Paris 11.5, Milan 10.6 and Madrid 9.6. Zurich and London are not covered, so each uses the median of the other fourteen cities, 13.45. The three travel means are Vienna 32 minutes from the Vienna Chamber of Labour commuter study, Tokyo 47.5 minutes from the 2021 Japanese Survey on Time Use and Leisure Activities (95 minutes of commuting and school travel per weekday in Tokyo, halved for a one-way estimate), and Warsaw 27 minutes from its school-travel survey [39]; each mean enters as $[\exp(-t/30), 1]$. These 11 fills are modelling proxies based on national or survey-level information; Section 5.5 tests how the ranking responds to the inputs most open to question.

All 160 city–indicator cells now have an assigned value or range: 123 are single values, including proxies, and 37 are known ranges; none is fully unknown. Each range enters the model with utility bounds ℓ_{ij} and h_{ij}.

5 Establishing the baseline evaluation model

5.1 Determining indicator weights (AHP)

Step 1: Set up the hierarchy. Our goal is to choose a city suited to high-school life. The five categories in Section 4.2 form the next level, with two indicators under each. We use the analytic hierarchy process (AHP) to turn our stated priorities into weights [2]. These judgements describe our student viewpoint.

Step 2: Compare the five categories. We use Saaty's 1–9 scale: 1 means equal importance, 3 moderate preference, 5 strong preference, 7 very strong preference and 9 extreme preference; 2, 4, 6 and 8 are intermediate values. Each entry in Table 3 compares its row with its

column, and the reverse comparison is its reciprocal. A teenager's safety and schooling come first, with equal importance. Cost matters through the family budget and carries the same weight as daily mobility. Broadband and pools are nice to have, so the daily-life category receives less weight.

Table 3: Pairwise category comparisons from our student viewpoint. S: safety and environment; C: cost of living; E: education; M: mobility; D: daily life.

Category	S	C	E	M	D
Safety and environment	1	3	1	3	4
Cost of living	$1/3$	1	$1/3$	1	2
Education	1	3	1	3	4
Mobility	$1/3$	1	$1/3$	1	2
Daily life	$1/4$	$1/2$	$1/4$	$1/2$	1

Step 3: Calculate category weights and check consistency. We find the principal eigenvector of the comparison matrix:

$$Av = \lambda_{\max} v. \tag{7}$$

Here A is the five-category matrix, $\lambda_{\max}$ its largest eigenvalue, and v the corresponding positive eigenvector.

The category weights are

$$g_k = \frac{v_k}{\sum_{h=1}^{5} v_h}. \tag{8}$$

Here g_k is category k's weight, and v_k is its eigenvector component.

We check how closely the comparisons agree using the consistency index:

$$CI = \frac{\lambda_{\max} - n}{n - 1}. \tag{9}$$

Here $n = 5$ is the number of categories and CI is the consistency index.

The consistency ratio is

$$CR = \frac{CI}{RI}. \tag{10}$$

Here $RI = 1.12$ is the random index for a five-category matrix.

The matrix gives $\lambda_{\max} = 5.0264$, $CI = 0.00660$ and $CR = 0.0059 < 0.1$, which meets the AHP consistency threshold [2].

Step 4: Compare the two indicators within each category. Each local comparison has the form

$$B = \begin{pmatrix} 1 & b \\ 1/b & 1 \end{pmatrix}. \tag{11}$$

Here B compares the two indicators, and b is the importance of the first relative to the second.

We give homicide twice the importance of $PM_{2.5}$, since immediate personal safety is our first concern. Living costs and housing are equally important parts of the family budget. Teacher resources receive three times the weight of higher-education supply because they concern current schooling. Travel time receives twice the weight of stops because it measures time spent on a journey. Broadband receives twice the weight of pools because students use it for schoolwork. Each two-indicator matrix is internally consistent.

The final indicator weight combines its category and local weights:

$$w_j = g_k p_j. \tag{12}$$

Here p_j is indicator j's local eigenvector weight within category k, and w_j is its global weight.

Table 4 gives all ten weights. Calculations use unrounded values; the displayed global weights sum to 1.000.

Table 4: AHP weights at the category and indicator levels, rounded to three decimals.

Category	Indicator	Category weight	Local weight	Global weight
Safety and environment	Homicide rate	0.340	0.667	0.227
	$PM_{2.5}$	0.340	0.333	0.113
Cost of living	Living-cost index	0.123	0.500	0.062
	Housing price-to-income	0.123	0.500	0.062
Education	Pupils per teacher	0.340	0.750	0.255
	Higher education	0.340	0.250	0.085
Mobility	Travel time	0.123	0.667	0.082
	Transport stops	0.123	0.333	0.041
Daily life	Broadband	0.073	0.667	0.049
	Swimming pools	0.073	0.333	0.024

Equal weights (0.1 each) are kept only as a comparison in the robustness section. For later preference comparisons, a multiplier adjusts each weight before renormalisation:

$$w'_j = \frac{m_j w_j}{\sum_k m_k w_k}. \tag{13}$$

Here $m_j > 0$ is a preference multiplier, and w'_j is the adjusted weight.

5.2 TOPSIS scoring

Step 1: Form the utility matrix. Each entry u_{ij} is city i's utility for indicator j, calculated in Section 4.2. All entries share a 0–1 scale, so no extra normalisation is needed. TOPSIS compares each city with an ideal profile of all ones and a worst profile of all zeros [3].

Step 2: Find the distance to the ideal.

$$D_i^+ = \sqrt{\sum_{j=1}^{10} w_j(1-u_{ij})^2}. \tag{14}$$

Here D_i^+ is city i's distance to the ideal, w_j is indicator j's weight, and u_{ij} is its utility.

A smaller distance means fewer weighted shortfalls from the ideal.

Step 3: Find the distance to the worst profile.

$$D_i^- = \sqrt{\sum_{j=1}^{10} w_j u_{ij}^2}. \tag{15}$$

Here D_i^- is city i's distance from the all-zero profile, using the same utilities and weights.

A larger distance means stronger performance above the worst profile.

Step 4: Calculate the score.

$$S_i = 100\frac{D_i^-}{D_i^+ + D_i^-}. \tag{16}$$

Here S_i is city i's score on a scale from 0 to 100, with higher scores indicating a better overall profile.

Each weight enters the squared distances once.

Step 5: Carry ranges into the score. The score rises with every utility, so the vectors of lower and upper utilities give the city's score range $[S(\ell_i), S(h_i)] = [L_i, U_i]$. For the main ranking, we calculate one representative score at the middle of each utility range. The full score range is used only in the robustness section.

5.3 Scoring results and interpretation

Zurich leads the AHP ranking with a representative score of 64.5. Madrid and Singapore follow at 63.3 and 62.5 (Table 5). The three tiers separate the leading group from a close middle and a lower cluster; Figure 2 shows the same pattern through category scores.

Top tier. Zurich, Madrid and Singapore score 62.5–64.5. Zurich's safety and environment score is 0.75, its mobility score is 0.70, and its daily life score is 0.78. Madrid combines 0.73 for safety and environment with 0.59 for education, while Singapore reaches 0.81 for safety and environment and 0.80 for daily life. Their advantage is a broad profile across the categories that carry the largest AHP weights.

Middle tier. Milan through Oslo score 57.6–61.1, with a 1.4-point gap from Singapore to Milan. Milan has education 0.59 and mobility 0.63; Dublin has safety and environment 0.64 and education 0.59; Paris has education 0.63 and daily life 0.72. Helsinki, Prague and Oslo each have a strong category, with mobility at 0.69 for Prague and 0.72 for Oslo, but their other categories keep the total below the top tier.

Table 5: City ranking under AHP weights. Scores use the middle of each known utility range.

Rank	City	Score	Tier	Rank	City	Score	Tier
1	**Zurich**	**64.5**	**Top**	9	Oslo	57.6	Middle
2	Madrid	63.3	Top	10	London	56.1	Bottom
3	Singapore	62.5	Top	11	Berlin	55.6	Bottom
4	Milan	61.1	Middle	12	Stockholm	55.4	Bottom
5	Dublin	60.8	Middle	13	Warsaw	54.5	Bottom
6	Paris	60.8	Middle	14	Vienna	54.4	Bottom
7	Helsinki	58.7	Middle	15	Tokyo	54.0	Bottom
8	Prague	58.4	Middle	16	Amsterdam	45.2	Bottom

Bottom tier. London through Amsterdam score 45.2–56.1. Vienna's education score is 0.63 despite a 0.41 safety and environment score, and Tokyo's education and mobility scores are 0.49 and 0.57. Amsterdam's safety and environment score is 0.24, driven by a homicide utility of 0.06, leaving it 9.0 points behind Tokyo. The tier reflects the balance across categories rather than one universal weakness.

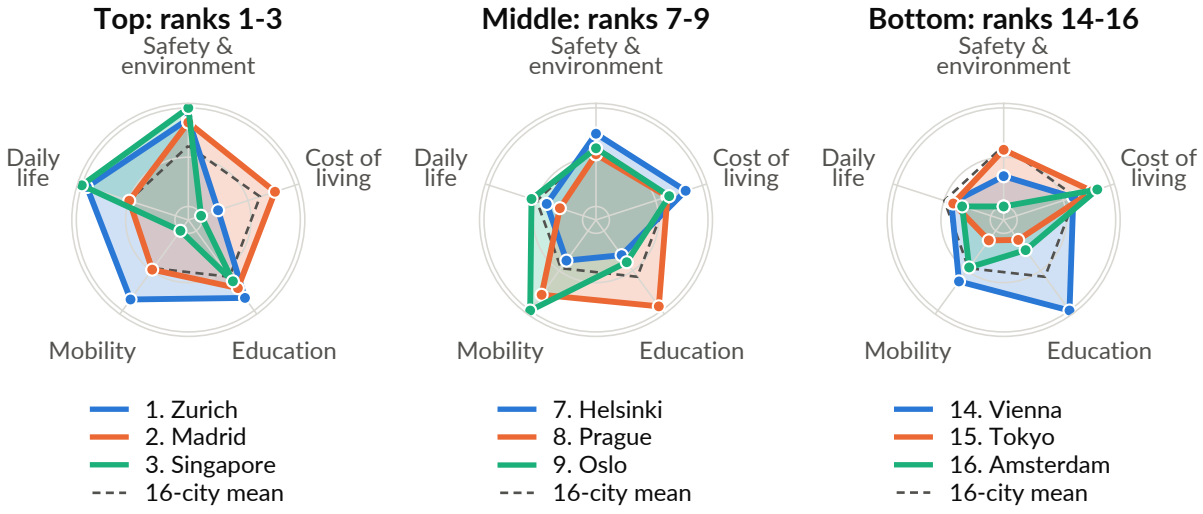


Figure 2: Category profiles of ranks 1–3, 7–9 and 14–16. Each axis is stretched from the lowest city (centre) to the highest city (outer ring) among all sixteen, and the dashed line is the sixteen-city mean. Category scores combine the two indicators in each category with the AHP weights in Table 4.

5.4 A significant revision: from equal weights to AHP weights

Equal weights provide a transparent baseline, but they give a swimming pool and homicide rate the same influence as the education and safety conditions that shape a student's daily life. The equal-weight ranking puts Zurich, Oslo, Paris and Dublin at the top. It therefore treats the ten indicators as equally important even though the decision is framed around high-school life.

We replaced that baseline with an analytic hierarchy process (AHP) elicited from a high-schooler's point of view. Safety and environment and education each receive category weight 0.340, cost and mobility receive 0.123 each, and daily life receives 0.073. Within categories, the largest local priorities are homicide rate (0.667 of safety and environment), pupil–teacher ratio (0.750 of education), and travel time (0.667 of mobility). The consistency ratio is 0.0059, so the pairwise priorities are internally consistent.

The comparison uses the same utilities, anchors and scoring formula under both weight sets. AHP ranks Zurich, Madrid, Singapore and Milan first; equal weights rank Zurich, Oslo, Paris and Dublin first. Twelve of sixteen cities change rank, and Kendall's τ is 0.517. Zurich remains first under both inputs, while the rest of the top group changes because AHP gives more influence to school resources, safety and the daily conditions around them.

5.5 Robustness of the recommendation

Score ranges and worst-case regret. Allowing every uncertain utility to move to either endpoint gives the score interval

$$[L_i, U_i] = [S(\ell_i), S(h_i)], \tag{17}$$

where ℓ_i and h_i are the lower and upper utility vectors for city i. For Zurich, Madrid and Singapore these intervals are 60.5–67.2, 58.5–66.6 and 59.4–65.1. The largest possible shortfall of city i behind any rival is

$$R_i = \max\left\{0, \max_{k \neq i} U_k - L_i\right\}. \tag{18}$$

Minimizing this regret still selects Zurich (6.1), followed by Singapore (7.8) and Madrid (8.7). Nine of sixteen cities can be first under some endpoint combination, so the representative ranking is a useful ordering of the shared inputs while the regret result gives the cautious recommendation.

Dependence on homicide inputs. Zurich remains first until its homicide rate reaches about 0.69 per 100,000, where Madrid draws level. Removing the Zurich homicide entry moves Zurich to eighth and makes Madrid first. Amsterdam's homicide utility is 0.06 and drives its last-place score; the supplied numerator is not reproduced by official CBS counts, and removing that entry lifts Amsterdam only to fifteenth [40].

Data withdrawals. Replacing selected point cells by the full range $[0, 1]$ gives the following first city under each weighting scheme.

Table 6: First city after withdrawing dated or flagged point cells.

Withdrawn group	Cells	First (AHP)	First (equal weights)
Before 2020	13	Zurich	Zurich
After 2025	64	Madrid	Zurich
Non-`ok` flags	28	Milan	Zurich
Union of the three groups	95	Milan	Milan

The retained 123 single values and 37 known ranges give Zurich under both schemes. All 31 controlled source cases from the earlier source audit also keep Zurich first, so the recommendation does not depend on one named source refinement.

Weights and preferences. Around the AHP weights, draw

$$w \sim \mathrm{Dirichlet}(\kappa w_{\mathrm{AHP}}).$$

Across 10,000 draws, Zurich is first in 56.3% at $\kappa = 20$, 82.8% at $\kappa = 100$, and 98.7% at $\kappa = 500$; Madrid takes most of the remaining first places. Doubling both education weights leaves Zurich first with score 63.3, ahead of Madrid at 62.1 and Singapore at 61.3. Doubling both mobility weights also leaves Zurich first.

VIKOR method check. Using the same utilities and AHP weights, we apply VIKOR with $v = 0.5$ in five steps.

1. For each city, compute the weighted group utility

$$S_i^{\mathrm{V}} = \sum_j w_j \frac{f_j^* - f_{ij}}{f_j^* - f_j^-}, \tag{19}$$

   where f_j^* and f_j^- are the best and worst utilities for indicator j.

2. Record the largest weighted criterion regret

$$R_i^{\mathrm{V}} = \max_j \left[w_j \frac{f_j^* - f_{ij}}{f_j^* - f_j^-} \right]. \tag{20}$$

3. Combine the two measures with

$$Q_i = v \frac{S_i^{\mathrm{V}} - S^*}{S^- - S^*} + (1 - v) \frac{R_i^{\mathrm{V}} - R^*}{R^- - R^*}, \qquad v = 0.5. \tag{21}$$

4. Order cities by increasing Q_i.

5. Check the acceptable-advantage and stability conditions for the first city.

VIKOR orders Madrid, Zurich and Paris first. The acceptable-advantage condition fails while the stability condition holds, so VIKOR does not separate Madrid and Zurich clearly. The method check supports reporting Zurich as the AHP-TOPSIS leader with a close Madrid alternative rather than treating the first rank as a large margin.

6 Model revision and extension

6.1 Looking ahead to 2035

A high-school student may live with a city's schools, services and travel for several years. We test three possible 2035 changes to see whether they alter city scores; these are conditional comparisons, not forecasts.

S1 raises higher-education institutions, transit stops and pools by 10%, as growing cities may expand public services. S2 raises living costs, the housing price-to-income ratio and travel time by 10% as housing demand and congestion grow. S3 lowers pupils per teacher, $PM_{2.5}$ and travel time by 10% through possible investment in schools, cleaner air and faster transit.

For each affected raw value x, we multiply by $r = 1.10$ or $r = 0.90$ and then recalculate utility with the same formulas and fixed anchors as Section 4.2; the resulting utilities go through the existing TOPSIS scoring model.

Zurich, Madrid and Singapore remain first, second and third. S1 adds 0.1–0.3 points, S2 subtracts 0.4–0.5 points, and S3 adds about 1.0 point; only Dublin and Paris swap fifth and sixth under S3. Ranks 7–16 keep their order in all three scenarios. Figure 3 shows the top six in each case.

Alternative priorities and accepted data can affect the ranking. Section 5.5 compares weight choices and tests withdrawals of selected source groups.

6.2 Within-city fairness and growth stages

Citywide supply per resident is not the same as student access. Fees, opening hours, transport and location affect which facilities a student can use. Bruno et al.'s grids place 99.18% of retained Oslo residents and 99.97% of retained Zurich residents within a modelled mean walk of 30 minutes to 20 learning POIs [60]. These resident-based shares provide context outside the scores.

District comparisons can show how school resources are distributed. For district pupil counts n_d and pupils per teacher r_d, the pupil-weighted utility is

$$\bar{u} = \frac{\sum_d n_d \, 15/(15 + r_d)}{\sum_d n_d}. \tag{22}$$

This gives each pupil equal weight when combining district staffing utilities.

Student travel can vary substantially within a city. For Istanbul's 2012 high-school cohort aged 14–18, the sharp minimum gap between overall mean travel utility and the mean for the lowest 20% is 0.226686 under the cohort's aggregate-data rules [17]. In Oslo's 2023 youth survey, 66% of grade 8 and 41% of Vg3 respondents rated local youth meeting-place provision very or fairly good [62]. Age-specific comparisons would need separate data for 14- and 18-year-olds.

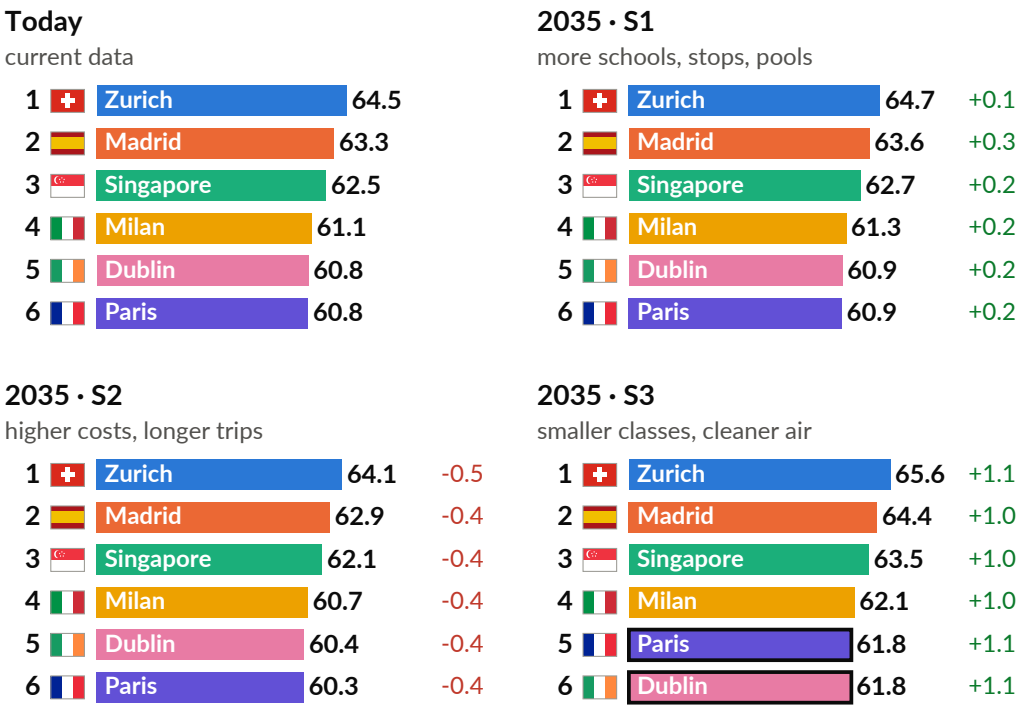


Figure 3: Top six cities today and in 2035 under each scenario. Bars show the TOPSIS score; numbers on the right give the change from today, and a box marks a city whose rank changed. S1: 10% more higher-education institutions, transit stops and pools. S2: 10% higher living costs, housing price-to-income ratio and travel time. S3: 10% fewer pupils per teacher, lower $PM_{2.5}$ and shorter travel time.

7 Comparing cities within the same country

The primary sample has one city per country. Frozen US and India cases use the same ten indicators and scoring formula, outside the sixteen-city ranking [21].

Phoenix has higher retained affordability utilities than New York. Mumbai has the higher $PM_{2.5}$ utility; Delhi leads in living-cost, housing, school-resource and broadband utilities. Both score pairs overlap. These historical city and education scopes identify neither a necessary winner nor a national causal effect.

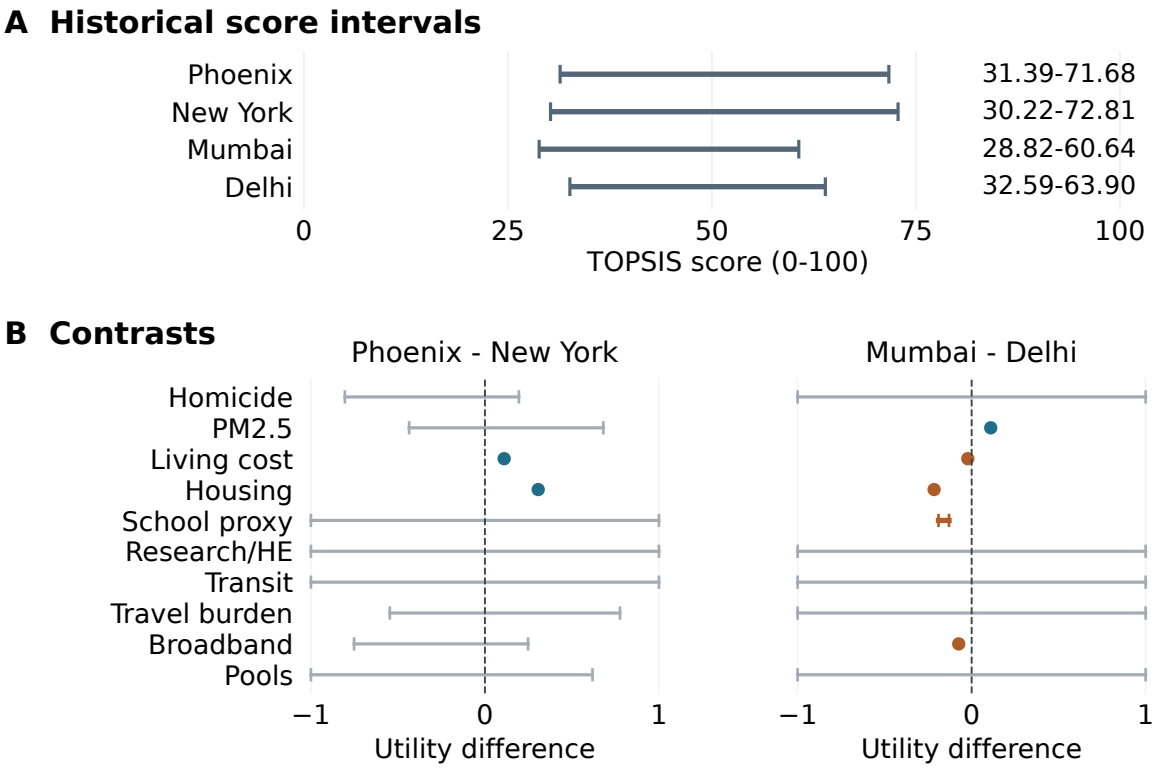


Figure 4: Frozen auxiliary V117 country cases, separate from the primary sixteen cities. Top: retained score intervals. Bottom: all twenty utility differences, first city minus second; seven exclude zero. Positive differences favour the first city; negative differences favour the second. Grey ranges leave direction unresolved. School resource retains 2024–25 school-location all-age XI–XII counts and nominal teacher-headcount allocation, not FTE. Living cost and housing use 2025 Numbeo aggregates; air uses 2021 WHO monitoring means; broadband uses the September 2026 index (July–September test window). Source years, boundaries and populations differ. Ranges are conditional identification bounds, not confidence intervals; neither pair identifies a necessary overall winner [21, 22, 23].

8 Conclusion and decision use

We recommend Zurich for these sixteen cities. Under AHP weights, its TOPSIS score is 64.5, ahead of Madrid (63.3) and Singapore (62.5). Safety and education each carry 0.340 weight; Zurich scores 0.749 and 0.605 in these categories, while mobility (0.700) and daily life (0.783) balance a lower cost score (0.445). Zurich has the smallest worst-case regret (6.1 points) and stays first in all 31 source cases and 82.8% of 10,000 weight draws at concentration 100. Removing its homicide figure makes Madrid first. The three 2035 scenarios keep the top three unchanged.

A city choice built around a student's day

Zurich ranks first among the sixteen cities with a score of 64.5, ahead of Madrid (63.3) and Singapore (62.5). Our team set the weights from a high-schooler's viewpoint; they are not survey results. Zurich has the smallest worst-case gap, stays first in all 31 source checks, and leads in about 83% of 10,000 weight trials. Removing its homicide figure makes Madrid first.

How the comparison works

The measures cover safety, air quality, housing and living costs, teaching resources, higher education, transport stops, journey time, broadband and swimming facilities. Each uses the same scale, with more weight for conditions the team judged important to a student. Safety and education each receive 34% of the category weight; cost and mobility receive 12.3% each, and daily life 7.3%. Of 160 city-measure entries, 123 contain a single value, including proxies, and 37 retain a known range.

The main score uses each range's midpoint. A separate check uses their least favorable ends; Zurich has the smallest possible gap to a rival. Equal weights give pools the same influence as classroom resources, while our priorities assign 2.4% to pools and 25.5% to pupils per teacher.

What new records change

Eleven entries use stated proxies in place of city-specific records. Eurostat's 2024 upper-secondary table `educ_uoe_perp04` supplies teacher ratios for six cities; Zurich and London use the median of the other fourteen. The Vienna Chamber of Labour commuter study, Japan's 2021 Survey on Time Use and Leisure Activities, and the Warsaw 2022 school-trip survey supply travel estimates [39]. Zurich's homicide rate is 0.45 per 100,000, based on two completed offences in the city's 2025 police statistics and a 2025 population of 440,935 [41]. Removing this input makes Madrid first.

Facility counts describe supply, not student access. A nearby stop still needs a suitable route, service time, fare and accessible connection; pool totals do not describe fees, opening hours or neighbourhood reach. Families can check these details with school options, budgets and daily routines.

Trying different priorities

The three 2035 scenarios apply 10% changes to selected measures. Zurich, Madrid and Singapore stay in the top three; only Dublin and Paris switch places elsewhere. These are conditional tests, not forecasts. Families can also compare schooling and travel priorities: when we doubled both schooling weights or both travel weights, Zurich remained first.

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Report on Use of AI Tools

Documentation status: incomplete review record. This appendix records assistance supported by saved task metadata and research receipts. The complete team AI history remains to be reconciled with actual saved conversations and other tool records before submission.

Known tool and scope

OpenAI Codex, used 8–11 November 2026 (Singapore time). The session header records Desktop version `0.166.0-alpha.5.1`; three inspected windows record `gpt-6.1-sol` in 74, 42 and 51 root contexts. The two continuation windows record 148 subagent spawn requests: 77 specify `gpt-6.1-terra`, 28 specify `gpt-6-luna`, and 43 omit a model or use null. Recorded identifiers and requested labels do not attest serving versions or every assistant's model. The assistant supported official-source search and extraction, data-scope checks, Python calculations and audits, LaTeX drafting, and wording revisions. Concurrent assistant tasks prepared bounded evidence checks, a controlled model diagnostic and the news-article contribution [19]. The anti-defensive-writing skill supplied editorial instructions; it is a workflow resource, not a surveyed student preference or a data source.

Exa, used 8–9 November 2026 through the recorded `web_fetch_exa` tool, retrieved and extracted text from official webpages, PDFs and data interfaces [20]. Saved receipts identify the service, source URLs and returned text, but do not specify an internal model or service version. They do not establish whether a generative model participated in every extraction. Extracted representations are distinguished from original downloaded bytes. For example, Tokyo sports-facility PDF text with ambiguous row order was excluded; the admitted five-venue lower bound uses separately downloaded Nerima municipal HTML.

Recovered records and coverage

The supporting inventory authenticates a complete-line prefix of this root task log by byte count and SHA-256. It ends at 14:31:18.850 UTC on 9 November 2026 (22:31 in Singapore). A private export retains 326 selected visible text records: six selected genuine requests, 259 progress messages and 61 complete final responses. The original 125-record selection and its private byte prefix are preserved.

The original tool-input export retains 1,865 plaintext inputs: 1,749 custom-call strings and 116 function-call argument strings, including generated code and LaTeX where present. Recorded inputs do not prove successful execution.

A continuation of the same root log ends at 07:26:15.500 UTC on 10 November 2026. It retains 197 selected visible records and 1,372 top-level plaintext assistant tool-call inputs. A third window ends at 20:32:42.716 UTC that day (04:32 on 11 November in Singapore), adding 239 visible records: three requests or replies, 191 progress messages and 45 final responses; it also retains 1,367 tool-call inputs. Across the three windows, totals are 762 selected visible records and 4,604 inputs. Nested executions, subagent sessions and later work are outside these selections.

All three windows exclude reasoning, system/developer records, tool results, attachments, subagent sessions, other chats and team members. The original log and selected exports remain outside Git (directory mode 700, file mode 600); these selections are not submission transcripts. Tool inputs remain unredacted and may contain copied content. Public inventories retain metadata and hashes; older selectors contain request prefixes, while the latest uses fingerprints. Record exclusion does not redact content, and fingerprints do not guarantee anonymity.

Verification associated with this draft

The research calculations preserve the original city CSV and the V205 primary evidence. The V205 validation independently recomputes weighted norms and self-excluded competitor maxima, verifies interval membership of constructed completions and checks source hashes. Original provider records are kept separately from derived tables. Compilation and rendered-page review check the presentation. These checks establish the specified calculation and document properties; they do not replace the team's responsibility to review factual claims, original supplied data and final authorship.

Records required to finalize this report

Finalization requires the team's actual AI tools, available models or versions, dates and purposes, with associated text and references, and saved query/response records according to the applicable COMAP policy and its exceptions. Other members' tools, translation use and earlier conversations remain unconfirmed. The registered Team Control Number and confirmation of complete team-use coverage are also required.